

ENVS 193DS Final

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[GitHub Repo](#)

Set Up

```
# loading in packages
library(tidyverse) # general use
library(janitor) # cleaning data
library(here) # organizing files
library(rstatix) # statistical relationships
library(MuMIn) # modelling
library(DHARMA) # residual diagnostic simulation
library(ggeffects) # model predictions
library(ggplot2) # visualizations

# read in data
```

```
nests <- read_csv(here("data", "nest_data_final.csv")) # read in and make object for nest data
dinner <- read_csv(here("data", "Dinner_Data.csv")) # read in and make object for personal data
```

Problem 1. Research Writing

a. Transparent Statistical Methods

In part 1, the researchers used a Generalized Linear Model (GLM) to reach their conclusion and compute that p-value, since the claim answers the central question of how distance to water edge predicts the probability of American Avocet microhabitat use. Additionally, since the variables are binary (use of microhabitat or no use) and continuous (distance from water edge) are non-parametric. Furthermore, since the response variable is binary we know the test was a logistic regression, a specific type of generalized linear regression. In part 2, they used a chi-square test because they are examining the relationship between two categorical variables, bird species and habitat type.

b. Suggestions for rewriting

Part 1.

We found that American Avocets do not use microhabitats in a random fashion, but rather with respect to distance to water edge. In this sense, distance to a water edge acts as a predictor for the likelihood American Avocet use a microhabitat.

Part 2.

When looking at the distribution of American Avocet, Forster's Tern, and Black-necked Stilt, we found that each species was not spread randomly among habitats. Rather, each species presence differed based on habitat. This conclusion takes into consideration managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflat as habitats.

c. Further Analysis

Another variable that could influence the probability of American Avocet microhabitat use is the predator abundance, which could be count data or categorical with pre determined levels from low presence to high presence of predators. This variable will influence American Avocet habitat use because this species might avoid areas with high concentrations of predators as they pose a risk for injury, mortality, and reduce reproductive success.

Problem 2. Data Analysis

a. Clean and Wrangle

```
# create new object, nest_clean
nests_clean <- nests |>
  # keep only height, bird species, and ant species
  select(height_cm, case_control, ant_species) |>
  # filter to include on C. mimosae, C. sjostedti, and C. nigriceps
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # order the species
  mutate(ant_species =
    as_factor(ant_species),
    ant_species = fct_relevel(ant_species,
                              "AB",
                              "RRB",
                              "BBR")) |>

# create new column for scientific species names
mutate(scientific_name = case_match(
  ant_species,
  "AB" ~ "Crematogaster sjostedti", #C. sjostedti
  "RRB" ~ "Crematogaster mimosae", # C. mimosae
  "BBR" ~ "Crematogaster nigriceps" # C. nigriceps
))

# display new structure of the data
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ scientific_name: chr [1:270] "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimosae" ...
```

```
# display 10 random rows from data
slice_sample(nests_clean, n =10)
```

```
# A tibble: 10 x 4
  height_cm case_control ant_species scientific_name
    <dbl>         <dbl> <fct>         <chr>
1  392         1       AB      "Crematogaster sjostedti"
2  310         0       RRB      "Crematogaster mimosae"
3  513         0       RRB      "Crematogaster mimosae"
4  154         0       RRB      "Crematogaster mimosae"
5  324         1       BBR      "Crematogaster nigriceps"
6  154         0       RRB      "Crematogaster mimosae"
7  324         0       RRB      "Crematogaster mimosae"
8  324         0       RRB      "Crematogaster mimosae"
9  324         0       RRB      "Crematogaster mimosae"
10 324         0       RRB      "Crematogaster mimosae"
```

1	267	1 RRB	<i>Crematogaster mimosae</i>
2	132.	0 AB	<i>Crematogaster sjostedti</i>
3	195	1 RRB	<i>Crematogaster mimosae</i>
4	185	0 RRB	<i>Crematogaster mimosae</i>
5	120	0 RRB	<i>Crematogaster mimosae</i>
6	103	0 RRB	<i>Crematogaster mimosae</i>
7	180.	1 RRB	<i>Crematogaster mimosae</i>
8	522	0 RRB	<i>Crematogaster mimosae</i>
9	157	0 RRB	<i>Crematogaster mimosae</i>
10	112	0 BBR	<i>Crematogaster nigriceps</i>